

Draft - expert review pending

This unit is complete and passes the platform's automated checks, but no reviewer has read it yet. Treat its sources and claims with care.

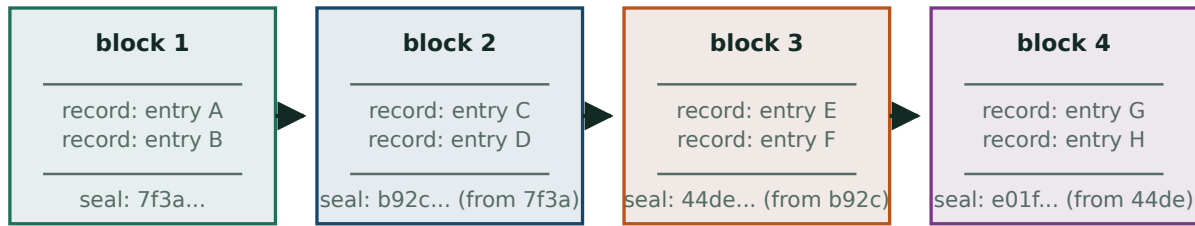
Topic 6.4 - Blockchain, Virtual Reality, and Augmented Reality

A real classroom situation

A district office announces that teachers' professional-development records will now be kept on "the blockchain", and a vendor offers a school a "VR lab" - four headsets and a demo reel. The head teacher asks the B.Ed trainee on practicum what these things are, and the trainee realises she has only slogans: unbreakable, immersive, the future. She cannot say what a blockchain is, whether records need one, or what the difference is between virtual and augmented reality - and a purchase decision worth a year's budget is about to rest on slogans. This topic replaces the slogans with one-paragraph understandings, which is what a teacher actually needs to advise a school.

A blockchain: blocks, seals, chain

Each block carries a seal derived from the previous block's contents



Change one old record and its seal changes -
which breaks every later seal, and every copy-holder can see the break.

"Blockchain" names this ledger design; bitcoin is one application of it, not the design itself.
The honest question: does this record need one? Whose trust problem does it have?

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Explanation

Blockchain

A blockchain is a record-keeping design in which entries are grouped into blocks, and each block carries a mathematical seal derived from the previous block's contents - a chain. The consequence is the property that matters: altering an old entry changes its seal, which breaks the link to every later block, and everyone keeping a copy of the chain can see the break. NIST's technical overview describes it as a tamper-evident, distributed ledger: many copies, no single silent editor (Yaga et al., 2018). Its famous first use, cryptocurrencies such as Bitcoin, is one application of the design, not the design itself - "blockchain" and "bitcoin" are not the same word.

For a school, the honest question is not "is blockchain good?" but "does this record need one?" A blockchain earns its cost when many parties who do not fully trust each other must share one record they all need to resist silent change - supply chains, land registries, credential verification across institutions. A school's internal attendance register, kept by one office that the school itself trusts, usually does not need one; the district records that must survive any

single office's failure might. The trainee's answer to the head teacher begins: "it depends on whose trust problem the record has."

Virtual and Augmented Reality

Virtual reality (VR) replaces the world: a headset covers the eyes and ears, and the wearer stands inside a computer-generated environment - a virtual science lab, a historical city, the inside of a cell. Augmented reality (AR) adds to the world: the learner looks through a phone or special glasses at the real classroom, and digital objects appear within it - a beating heart floating above the desk, labels pinned to real equipment. VR immerses; AR annotates.

VR immerses; AR annotates

Property	Virtual reality	Augmented reality
What it does	replaces the world with a generated one	adds digital objects to the real one
What the learner needs	a headset and a safe clear space	a phone or tablet camera
One classroom use	walk through a virtual historical city	label the parts of real lab equipment
Main constraint	cost of devices and shared headsets	smaller screens and holding the device

For a Pakistani school, the realistic entry is AR on existing phones.
A purchase without a lesson plan is the thing to refuse.

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Reviews of AR and VR in education find real engagement gains - immersion holds attention, and simulation lets pupils visit what a school cannot physically provide (Al-Ansi et al., 2023) - alongside consistent constraints: devices are costly, content must match the curriculum, and evidence of learning gain beyond engagement is still maturing. For a Pakistani school, the realistic entry is AR on existing phones (a class set of phones is cheaper than a class set of headsets, and phones are already in most teachers' pockets), used for short, curriculum-anchored moments: label the skeleton, overlay the water cycle on the real sky. The four-headset "VR lab" of the opening story, with a demo reel

and no curriculum plan, is the purchase to refuse - not because VR is bad, but because a purchase without a lesson plan is.

Both technologies return to the unit's closing rule: the question is never "is it the future?" but "what does it do for this lesson, at this cost, with these cautions?" The trainee's briefing ends where the course does - with a teacher who can ask that question in one sentence and answer it in one page.

Activity: Slogan translation

Work in pairs. About 15 minutes.

1. Your teacher gives each pair five vendor slogans: "unbreakable blockchain records", "immersive VR learning", "AR brings lessons to life", "the school of the future", "tamper-proof certificates".
2. Translate each slogan into a claim that can be checked: what exactly would the product do, and what would we need to verify before buying?
3. For the blockchain-records claim, write the trust question: whose trust problem does a school record have, and does this solve it?
4. For the VR claim, write the two questions: what lesson, and what does it replace or add that a phone or a poster cannot?

Check your understanding

1. Define blockchain in one sentence and name its key property. (*Remember*)
2. Explain why "blockchain" and "bitcoin" are not the same word. (*Understand*)
3. A school's internal attendance register is proposed for blockchain. Analyse whether it needs one, using the trust question. (*Apply*)
4. State the difference between VR and AR with one classroom example of each. (*Understand*)

Summary

Blockchain is a chained, sealed, distributed ledger: old entries cannot be silently changed without breaking every later seal - valuable when distrusting parties

must share a resistant record, unnecessary when a single trusted office keeps its own. Virtual reality replaces the world with a generated one and needs headsets; augmented reality adds digital objects to the real one and runs on phones. Both show engagement gains and both demand the closing question: what does this do for this lesson, at this cost, with these cautions?

Self-assessment checklist

Sign in to sync your ticks across devices.

- ☐ I can define blockchain and its key property in plain language.
- ☐ I can apply the trust question to a proposed blockchain use.
- ☐ I can distinguish VR from AR with a classroom example of each.

Try this at your practicum school

Ask what technology purchases the school has made or been offered in the last two years, and what each actually gets used for now. Write two sentences: the pattern you notice, and the one question you would add to the school's purchase process.

Summative task

The head teacher has a fixed budget and three offers: a four-headset VR lab with demo content, an AR science pack running on phones the school already owns, and a blockchain certificate service for pupil records. In one page, assess each offer (what it is, what it would do, its cautions) and recommend one with conditions, using this topic's ideas and at least one rule from Units 4 or 5.

Marking guide (10 marks):

- Each offer explained accurately in plain language: 3 marks.
- Each assessment uses the right tool (trust question, lesson-fit question) rather than enthusiasm (Analyse): 4 marks.
- The recommendation carries conditions and a budget rationale: 2 marks.

- Fits one page: 1 mark.

Further reading

- Yaga, D., Mell, P., Roby, N., & Scarfone, K. (2018). *Blockchain Technology Overview* (NIST IR 8202). National Institute of Standards and Technology. <https://doi.org/10.6028/NIST.IR.8202>
- Al-Ansi, A. M., Al-Ansi, A. M., Jaboob, M. S., & Garad, A. (2023). Analyzing augmented reality (AR) and virtual reality (VR) recent development in education. *Social Sciences & Humanities Open*, 8, 100532. <https://doi.org/10.1016/j.ssaho.2023.100532>
- Stair, R., & Reynolds, G. *Principles of Information Systems*. Cengage Learning. (Course guide reading; print monograph, cited at bibliographic level.)

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